

Ege Ayhan

Computer Science Student · Software Developer

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SUMMARY

Computer Science student at İzmir Institute of Technology, with an Erasmus exchange in Germany. I build across the full stack — from React/Firebase web apps and PyTorch ML tooling to bare-metal C/C++ systems, including a custom audio-player OS for a real phone. I care about how software works underneath the framework, and I enjoy taking projects from idea to running code.

EDUCATION

B.Sc. Computer Science — İzmir Institute of Technology (IYTE)

2024 – 2028 (expected)

İzmir, Türkiye

Erasmus+ Exchange — Germany

Exchange

Coursework incl. Algorithm Engineering; German language (B1.1)

SELECTED PROJECTS

EgePod — Bare-metal Digital Audio Player OS

github.com/Afillex/EgePod

Custom OS for the Redmi Note 10S: three single-purpose C/C++ daemons (audio, power, UI) over Unix-socket IPC, a pure ALSA pipeline, and a 20–30 mA power budget. Ships with a simulator to run without the hardware.

C · C++ · ALSA · Linux kernel · POSIX IPC

GraphMaker — AI-agent proof visualizer

github.com/Afillex/GraphMaker

C++ state-machine engine driving a Node REST/WebSocket layer and a browser renderer (Cytoscape.js + KaTeX), letting LLM agents build step-by-step graph proofs.

C++ · Node.js · WebSocket · JavaScript

CNN Explorer — interactive deep-learning tool

github.com/Afillex/cnn-explorer

Design a convolutional network with sliders, train it live, and visualize feature maps and weight distributions — built with PyTorch and Streamlit.

Python · PyTorch · Streamlit

WG Wars — real-time multiplayer game

github.com/Afillex/wg-wars

Multiplayer bluff/bidding party game with room-code lobbies and timed rounds, synced in real time with Firebase. React 19 + Vite.

React · Vite · Firebase

minimal-api — HTTP server in pure C

github.com/Afillex/minimal-api

Dependency-free HTTP API built directly on POSIX sockets, with custom request parsing, routing, and per-endpoint handlers.

C · POSIX sockets · HTTP

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Java

Web: React, Next.js, Node.js, Vite, Firebase, REST/WebSocket

AI / ML: PyTorch, Streamlit, NumPy, pandas

Systems & Tools: Linux, ALSA/embedded, Make/CMake, Git, raylib

LANGUAGES

Turkish — Native English — Fluent German — B1.1 (improving)